Fiscal Year:	FY 2018	**Task Last Updated:**	FY 01/19/2018
PI Name:	Su, Ching-Hua Ph.D.		
Project Title:	Crystal Growth of II-VI Compound Semiconductors		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	256-544-7776
Organization Name:	NASA Marshall Space Flight Center		
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PI Web Page:			
City:	Huntsville	**State:**	AL
Zip Code:	35812	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	10/01/2006	**End Date:**	10/01/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Rogers, Jan	**Contact Phone:**	256.544.1081
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Flight Program:	ISS		
Flight Assignment:	NOTE: Changed end date to 09/30/2018, per PI (Ed., 10/18/16); note revised to 10/1/2018 to accommodate final reporting (Ed., 1/19/18) NOTE: Change in end date per P. Downey/MSFC (Ed., 9/28/2011)		
Key Personnel Changes/Previous PI:	FY2018 report: Prof. N. B. Singh, University of Maryland, Baltimore County and Prof. M. Dudley, Stony Brook University are supporting this project. Previous Co-Investigator Dr. Narayanan Ramachandran is no longer CoI on the project.		
COI Name (Institution):			
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Two reasons have been put forward to account for the improved crystal quality when crystals are grown in a microgravity environment. The first is weight-related reductions in crystal strain and defects. These are thought to be caused by the weight of the crystals during processing at elevated temperatures and retained on cooling, particularly for materials with low yield strength. The second and more general reason is related to the reduction in density-gradient driven convection in the fluid during crystal growth in reduced gravity. The objectives and experiment description for the proposed flight experiment “Growth of Ternary Compound Semiconductors” by (1) vapor growth and (2) melt growth are described below. 1a) Vapor Growth The physical vapor transport (PVT) crystal growth process consists of essentially three processes: sublimation of source material, transport of vapor species, and condensation of vapor species to form crystal. The latter two processes can be affected by the convection caused by gravitational accelerations on Earth. Reductions in such convection in low gravity is expected to yield a nearly diffusion-limited growth condition which results in more uniform growth rates (on the microscopic scale) and hence greater crystalline perfection and compositional homogeneity. This reduction of convective contamination in a reduced gravity environment will help to simplify the complexity of the coupled mass transport and growth kinetics problem. An improved comparison between the experimental results and theoretical simulations is also expected and will be beneficial to any growth process involving vapor mass transport or vapor-crystal interface growth kinetics. The materials to be studied in this project are ZnSe and the related ternary compound semiconductors such as ZnSeTe, ZnSeS, and ZnCdSe. The scientific objectives of the growth of ZnSe crystals in reduced gravity using physical vapor transport (PVT) processes are: • to establish the relative contributions of gravity-driven fluid flows to (i) the non-uniform incorporation of impurities and defects and (ii) the deviation from stoichiometry observed in the grown crystals as a result of buoyancy-driven convection and irregular fluid-flows caused by growth interface fluctuations and interface evolution. • to assess self-induced strain effects developed during processing at elevated temperatures and retained on cooling caused by the weight of the crystals from x-ray topography and by measuring the density and distribution of various crystalline defects. • to obtain high quality space-grown materials for various thermophysical and electrical properties measurements and as substrates for device fabrication and thus assess device performance as influenced by a substantial reduction in gravity-related effects. 1b) Melt Growth The melt growth of homogeneous ternary compound semiconductors, such as cadmium zinc telluride (Cd1-xZnxTe, $0 < x < 1$) is difficult because the thermal and compositional gradients in the melt just ahead of the growth interface during directional solidification can cause substantial gravity-driven fluid flows that alter the relative distribution of the two (CdTe and ZnTe) binary components and introduce structural defects into the grown crystals. The scientific objectives of the growth of CdZnTe crystals in reduced gravity using directional solidification process are: • to investigate quantitatively the influences of gravitationally dependent phenomena on the growth process and the incorporation of impurities and defects in the resulting crystals. • to better understand and control the growth parameters in the production process on Earth so as to increase the yield of usable material from the grown ingot. • to assess strain effects developed during processing at elevated temperatures and retained on cooling, caused by the weight of the crystals and, possibly, the contact with the growth container, from the structural characterization and by measuring the density and distribution of various crystalline defects. • to obtain high quality space-grown materials for various electrical and optical property measurements and as materials for the x-ray and gamma-ray detector fabrication as well as substrates for infrared device application and thus assess device performance as influenced by a substantial reduction in gravity-related effects.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	• In crystal growth there is a need to understand the relation between processes in the fluid phase (both liquid and vapor), such as buoyancy driven convection and the incorporation of impurities, and defects in the resulting crystal • Relation between fluid phase processes and the generation of defects in a grown crystal is an outstanding problem in materials growth • Crystal quality greatly influences important electronic properties in materials • Ternary compound semiconductors are of vital national interest as sensors in x-ray telescopes and for homeland security, and as substrate materials for infrared sensors • Studies in microgravity will be compared with modeling and will be used to optimize ground-based experiments
Task Progress:	The main disadvantages of the existing vapor growth techniques, such as physical vapor transport, are that (a) the significant amount of waste from starting material, and (b) the crystal growth rates are low and inconsistent which lowers the single crystal yield. In this period, progress has been made such that, with a simple extra procedure of the growth process, a much higher production yield can be consistently achieved and a routine production of high-quality single crystals of these semiconductors can be realized. During the processing of advanced electronic/optic materials at elevated temperature, the sample intend to interact with its fused silica container and form chemical bonding, or the so-called “wetting,” which makes the surface section of the sample attaching to the ampoule wall. When the sample is shrinking during cool down after the processing, the wetting area retains the attachment and causes internal cracks and other structural defects. These defects are detrimental to the electronic/optic performance of materials, especially on the applications of high-quality compound semiconductors. In this period, progress has been made that by adding a simple procedure, the wetting of the sample can be reduced during

	the high temperature processing. The effect of the innovation was tested on the growth of ZnSe and Fe-doped ZnSe semiconductors. A New Technology Report, 'A Method to Enhance Production Yield of Crystal Grown by Physical Vapor Transport Process,' was submitted as MFS-33528-1 on 6/07/2017.
Bibliography Type:	Description: (Last Updated: 02/18/2025)
Abstracts for Journals and Proceedings	Baird JK, Su C-H. "Acoustic effects in classical nucleation theory." Annual March Meeting of the American Physical Society, New Orleans, LA, March 13-17, 2017. Bulletin of the American Physical Society. 2017 Mar;2(4): V17.00009. http://meetings.aps.org/link/BAPS.2017.MAR.V17.9 , Mar-2017
Abstracts for Journals and Proceedings	Singh NB, Su C-H, Arnold B, Choa F-S, Bohorfous S. "Multifunctional 2D-Materials: Selenides and Halides." Presented as keynote talk in International Conference on Multifunctional Materials for Device Applications (ICMDA-2016), Panta, India, October 26-28, 2016. International Conference on Multifunctional Materials for Device Applications (ICMDA-2016), Panta, India, October 26-28, 2016. , Oct-2016
Articles in Peer-reviewed Journals	Su C-H. "Modification on thermoelectric properties of PbTe-based materials by incorporation of impurities." Current Smart Materials. 2017;2(2):82-92. https://doi.org/10.2174/2405465802666170322152909 , Feb-2017
Papers from Meeting Proceedings	Singh NB, Dayal V, Su C-H, Arnold B, Choa F-S, Kabandana GK, House D. "Ionizing organic compound based nanocomposites for efficient gamma-ray sensor." SPIE Optics-Photonics (International Society for Optics and Photonics) 2017, San Diego, CA, August 6-10, 2017. In: Proceedings SPIE Optics-Photonics, Volume 10392, Hard X-Ray, Gamma-Ray, and Neutron Detector Physics XIX; 1039210, September 2017. http://dx.doi.org/10.1117/12.2279121 , Sep-2017
Papers from Meeting Proceedings	Singh NB, Choa F-S, Su C-H, Arnold B, Kelly L. "Low Temperature Flux Growth of 2H-SiC and Beta-Gallium Oxide." Paper presented at International Semiconductor Device Research Symposium 2016 (ISDRS 2016), Bethesda, MD, December 7-9, 2016. International Semiconductor Device Research Symposium 2016 (ISDRS 2016), Bethesda, MD, December 7-9, 2016. , Dec-2016